Sole Source (Including Brand Name) Justification - Simplified Procedures for Certain Commercial Items

NOTE: *If a Justification was approved for the preceding acquisition, a copy of the approved Justification for the predecessor action must be included in the staff package for approval of the instant Justification. This applies to Justification staff packages that are submitted for review and approval at a level above the contracting officer. The predecessor Justification will be used as a reference document by the approving official.*

Is this a new or amended J&A Document? ☒ New ☐ Amended *(Prior to Award Only!)*

Is this a Bridge Action as defined in the AF Bridge Action Reduction Plan**?** ☐ Yes ☒ No

Funding level for this acquisition: [redacted]

Contracting Activity: Air Force Material Command
AFSC/PZIMC
Road A, Bldg 3705
Tinker AFB, OK 73145-3303

Purchase Request / Local ID Number: To Be Determined / [redacted]

Program / Project (and PE, if applicable): Next Gen High-Pressure Cold Spray System

Program Type (PEO or Other Contracting): Operational

Authority: RFO FAR 12.201-1, Simplified Procedures (41 U.S.C. 1901)

Estimated Contract Cost [redacted]

Justification Type: ☐ Class ☒ Individual

COORDINATION

Date	Project Lead / Program Mgr / Requiring Activity	
01 Apr 2026	[redacted]	[redacted]

APPROVAL

Date	Contracting Officer	
01 Apr 2026	Marc J. Kreienbrink AFSC/PZIMC; [redacted]	[redacted]

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I. Contracting Activity.

Department of the Air Force
Air Force Materiel Command (AFMC)
Maintenance Contracting Branch (AFSC/PZIMC)
[redacted]
Tinker AFB, OK 73145-3303
PR # to be determined
Next Gen High-Pressure Cold Spray System

II. Nature and/or description of the action being processed.

This is a Brand Name Justification for the award of a firm fixed price contract for a Next Gen High-Pressure Cold Spray System for the [redacted] 76th Propulsion Maintenance Group (76 PMXG), Oklahoma City Air Logistics Complex (OC-ALC), Tinker AFB, OK. This system and its components are commercial-off-the-shelf (COTS) items that are readily available from commercial suppliers.

[redacted]

III. Description of supplies/services required to meet agency needs.

76 PMXG requires a new cold spray system with capabilities to industrialize the cold spray repair technology and support engine sustainability. Equipment must be able to utilize Helium and Nitrogen carrier gas together within a mixed gas spray or individually in pure gas sprays, with capability to perform warm-up and cool-down functions with Nitrogen and production spray with helium. Equipment needs to be able to achieve 1,100C° applicator temperatures in order to effectively spray aluminum via Nitrogen carrier gas without the use of a second particle phase, i.e. pure Al powder. Equipment must have all required safety equipment installed prior to operation such as an Argon fire suppression system, and and blast handling systems added to the dust collector. Equipment must be installed with sensors capable of measuring evolving stress, particle speed, and particle density to support next generation coating and repair evaluation and quality control.

A new Nitrogen gas supply is required to provide the carrier gas for cold spray. The system must provide 105scfm of gas at 1000psig and 99.999% purity to the system. The purity requirement [redacted] as oxygen present in the carrier gas will erode and damage the heating elements in the system. High pressure gas plumbing shall be installed between the gas storage equipment and the cold spray booth.

The existing booth already has a dust handling system. However it is not designed to capture and handle reactive metals such as aluminum. The existing system must be modified with a spark arrester and blast venting devices to ensure safe operation.

All pieces of equipment required to meet this need are COTS items.

The jet engine production processes that the thermal spray system performs must proceed in accordance with the technical order for that engine. The technical order is the guidance and acceptance criteria for all work done on the thermal spray system.

IV. Statutory authority permitting other than full and open competition.

RFO FAR 12.201-1, Simplified Procedures (41 U.S.C. 1901)

V. Demonstration that the contractor's unique qualifications or nature of the acquisition requires the use of the authority cited above (applicability of authority).

Four pieces of equipment are uniquely capable of meeting the above requirements: the Impact Innovation EvoCSII cold spray machine, FANUC M-710iC/50 industrial robot, Oseir HiWatch HR2 spray sensor, and the ReliaCoat ICP-8 in situ stress sensor.

The Impact Innovations EvoCSII cold spray machine is the only cold spray machine on the market capable of producing alloyed aluminum coatings without the need to introduce a second phase into the coating. All replacement parts are manufactured to defined dimensions and newer iterations on existing parts are produced under unique part numbers to ensure consistency for prior repair development. The acquisition of an Impact EvoCSII system will allow the government to industrialize the cold spray process for repairs that involve hundreds or thousands of parts per year. [redacted] If the government cannot obtain Impact branded equipment it will end up with another [redacted] system that cannot meet the need [redacted] The acquisition of another VRC cold spray system rather than the Impact represents [redacted] VRC equipment [redacted] for need to the depot mission and the acquisition of another would prevent 76 PMXG from leveraging the technology to perform the kind and quantity of work inherent to depot maintenance.

The FANUC M-710iC/50 industrial robot is an exact match for the existing cold spray robot. This match allows for robot programs to be directly transferable between cold spray cells with minimal rework and ensures existing operators in the [redacted] are properly trained in its operation. Robot training in the depot is [redacted] The operators, technicians, and engineers at Tinker do not have knowledge of any other robot coding language other than FANUC. The introduction of another brand of robot would impose the need to train on an entirely different code language for the operation of a single cell and ensure that no operators from other robotic processes across the depot would qualify for jobs [redacted] Estimating the cost of this training burden is difficult because it expands beyond entry level operators to technicians, engineers and IT support staff. Technicians achieve their high level of competence through years of experience rather than simple coursework and this cannot be duplicated easily. This robot would also not fit into existing enterprise level management tools for the robot ecosystem that allow for program management, backups, health monitoring and other [redacted] industrial engineering tasks to maintain the mechanical health, reliability, [redacted] in the depot. If another brand of robot is acquired on this contract it would be considerably cheaper and faster for the government to scrap the new robot and execute another acquisition for a new FANUC robot than it would to try to obtain proficiency in a new language.

The Osier HiWatch HR2 sensor is the sensor in use across all USAF cold spray cells and the competing sensors use a different physical phenomenon to attempt to measure particle speed and density. The use of different physical phenomenon to capture the data result in different biases in the measurements that are not clearly correlated between sensors. Until ongoing correlation efforts are completed the production environment cannot utilize equipment other than the HiWatch HR2 to compare results between installations. If another brand of sensor is acquired it will simply not be used. Tinker AFB does not have the capability or technical expertise to perform the level of research required to correlate the different measurement phenomenon to make data sets from the different tools useful together. If Tinker were to contract out this kind of research it would represent 3-5 years at minimum before the data from the new sensor was useful in comparing results from the new equipment to the rest of the data generated in the USAF, with no guarantee the research would find a useful correlation at all.

The ReliaCoat ICP-8 is the only sensor available on the market for the real time capture of in situ (in position) evolving and residual stress in thermally sprayed coatings. This technology fits into PMXG's efforts to modernize [redacted] operations, to include updating coating evaluation specifications, maintaining higher levels of process control, reducing part rework, decreasing poor spray performance, and [redacted] The ICP-8 sensor also gives [redacted] a way to simultaneously compare coating performance between the shop's varied Cold Spray systems. It is very difficult to compare equipment settings between machines from different manufacturers and their effect on the as-sprayed properties of the coating, and this in-situ sensor will give the [redacted] much more insight into how X set of machine parameters on the VRC Gen III system compare to Y set of machine parameters on the new Impact Innovations system, which will allow PMXG to optimize performance of each system.

VI. Description of efforts made to ensure that offers are solicited from as many potential sources as practicable.

The solicitation will be posted on SAM.gov in order to facilitate maximum competition IAW RFO FAR 5.2. A copy of this justification will be posted along with the solicitation.

VII. Determination by the Contracting Officer that the anticipated cost to the Government will be fair and reasonable.

Ideally, prices will be determined fair and reasonable though adequate price competition, which we expect for this requirement based on market research. In the event only one offer is received, the Contracting Officer will ensure that prices will be negotiated at a fair and reasonable price to the government through price analysis. Being COTS items, certified cost and pricing data cannot be requested; however, other than certified cost and pricing data and redacted invoices of similar items can be requested for comparison purposes. The Contracting Officer will request a technical evaluation from the SME to aid in determining acceptability of contractor offers.

VIII. Description of the market research conducted and the results, or a statement of the reasons market research was not conducted.

Market research was conducted through Internet Searches/Contacting Known

IX. Any other facts supporting the use of Other Than Full and Open Competition.

The Air Force is not limiting competition but rather limiting the award to a specific brand name, where competition is anticipated based on thorough market research.

X. List of any sources that expressed, in writing, an interest in the acquisition.

[redacted]

[redacted]

XI. A statement of the actions, if any, the agency may take to remove or overcome any barriers to competition before making subsequent acquisitions for the supplies or services required.

In an effort to identity and remove barriers to competition, the Government will continue to monitor the marketplace for additional products with would meet the Government's needs. Approval of this justification does not relieve the ongoing responsibility of AFSC/PZIMC to actively assess and pursue competition.

Throughout this contract's PoP, the USAF intends to foster future competition by occasionally reviewing and challenging asserted data restrictions. Furthermore, should other source(s) become known during the life of this contract, full and open competition will be utilized to the maximum extent practicable. However, no further action will be taken at this time to attempt to acquire the necessary technical manufacturing and engineering information required to draft qualification requirements as all market research confirmed the OEMs are the only sources capable of meeting the USAF's requirements.

XII. Certification by the Contracting Officer.

As evidenced by my signature above, I have determined this document to be both accurate and complete to the best of my knowledge and belief.

XIII. Certification by the technical/requirements personnel.

As evidenced by my (our) signature(s) above, I (we) certify that any supporting data contained herein, which is my (our) responsibility, is both accurate and complete.